TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to present a new research and development initiative to build a causal world model for robotics systems. The proposed technology leverages score-based causal representation learning (CRL), combining interventional, multi-view, and temporal data to learn identifiable latent state and dynamics from high-dimensional sensor observations. This Technology Proposal follows the structure of the 2026 TIPA International Joint R&D Program template and is intended to facilitate collaboration with Korean SMEs for joint R&D and technology transfer.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**CausalWorld**: Causal World Model for Robotics			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (), AI (X), Next-gen Communication ( ), Advanced Robotics & Manufacturing (X), Quantum (			
Industrial Technology Classification [1)]	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Artificial Intelligence		Robotics & Automation	
	Medium Category	Small Category	Medium Category	Small Category
	Artificial Intelligence Machine Learning Data	World models causal reasoning Generative models	Robotics	Manipulation Autonomy Perception & planning
Concept of Proposed Technology	**Problem Statement and Market Need** Robotics systems increasingly rely on learned "world models" to predict the consequences of actions, plan safely, and adapt to new environments. However, conventional representation learning and predictive world modeling mainly capture correlations in training data and can fail			

when interventions (e.g., new control policies, payload changes, or novel environments) shift the data distribution. Strong generalization for robotics requires identifying the true low-dimensional latent structure that generates sensory observations and the cause–effect relationships among these latent variables.

A causal world model enables hypothetical (counterfactual) evaluation: answering "what if" questions by virtually testing interventions before executing them on hardware, reducing cost and risk in real-world experimentation.

Core Technology Description

CausalWorld is a modular, score-based causal representation learning (CRL) framework that learns a structured latent state and causal dynamics from high-dimensional robot observations (e.g., RGB images). It combines three key sources of statistical diversity to achieve identifiability and robustness:

1) Interventional CRL from actuation (label-free state discovery): Robots provide natural interventions by commanding actuators. Building on the ROPES methodology, we collect datasets where individual joints (or other controllable factors) are distributionally intervened upon, and learn an encoder that recovers controllable latent variables (e.g., joint angles, gripper width) from images without requiring per-image pose labels.
2) 2) Hybrid interventional + multi-view CRL (full observability and reduced sample complexity): To address partial observability (e.g., out-of-plane motion), CausalWorld uses multiple synchronized camera views and enforces score-consistency across both views and intervention environments. This hybrid approach strengthens constraints and improves identifiability even when views are noisy or partially entangled.
3) Temporal CRL for causal dynamics (predictive rollouts and planning): Robotic interaction data is sequential. CausalWorld learns causal dynamics in latent space by leveraging temporal variation as an implicit source of soft interventions, enabling recovery of time-evolving causal structure and state transitions. For continuous control, we incorporate continuous-time dynamics (e.g., neural ODE/SDE style models) in the latent space.

Once the latent state and causal mechanisms are learned, CausalWorld supports counterfactual rollouts by intervening directly on latent variables (e.g., holding a joint state fixed, modifying object mass/friction, or changing task goals) and predicting downstream effects. The resulting low-dimensional causal state can also serve as a compact interface to downstream planners and Vision-Language-Action policies, accelerating learning compared to planning directly in pixel space.

	Deployable Outcomes for Korean SME Partners Through a joint R&D project, the Korean SME partner will obtain a transfer-ready prototype toolkit (target TRL 4) including: • Data collection scripts for interventional and multi-view robotics datasets (simulation-first with a digital twin; extendable to real robots). • A training and evaluation pipeline for learning causal latent state, causal graphs, and temporal dynamics from RGB observations. • APIs for counterfactual (what-if) simulation in latent space, enabling safer and faster policy iteration. • Benchmarks, metrics, and documentation for partner adoption (e.g., pose/state estimation accuracy, OOD robustness, counterfactual validity, and policy transfer). • These outputs are designed for direct integration into partner robotics products and internal development workflows (ROS 2 / simulator integration).
Rationale for Joint R&D Project	**Why a Joint R&D Project Is Necessary** Building a deployable causal world model requires tight coupling between (i) theoretically principled CRL algorithms, (ii) realistic robotics data and interventions, and (iii) integration into end-to-end autonomy stacks. A joint project enables: • The research institute will translate recent score-based CRL advances into scalable software, validated on robotics data regimes (interventional, multi-view, and temporal). • The Korean SME partner will provide representative robot platforms, sensing configurations, and industrial task requirements, ensuring the resulting world model is deployable and commercially relevant. • Co-development of benchmarks, success metrics, and integration interfaces that reduce technology-transfer risk and accelerate productization.
Key Words	1. Causal world models for robotics 2. Causal representation learning (CRL) 3. Multi-view representation alignment 4. Temporal causal dynamics 5. Sim-to-real robust planning

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2)	**End of the Project**	Level (4)
R&D Objectives	**Project objective** • Develop and validate, at TRL 4, a causal world model prototype that learns identifiable robot-and-scene latent state and dynamics from high-dimensional observations (e.g., RGB images), enabling counterfactual prediction and planning with minimal labels. • Enables strong generalization (out-of-distribution robustness) by learning causal factors rather than spurious correlations. • Reduces reliance on expensive labeled datasets by exploiting interventions naturally			

	available in robotics (actuation). • Supports safer development through hypothetical evaluation (what-if rollouts) before executing interventions on hardware. **Specific R&D objectives** 1) Interventional causal state learning: Recover controllable latent variables (joint angles, gripper state, object pose) from images using score-based interventional CRL without per-sample labels. 2) Hybrid interventional + multi-view learning: Align latent state across multiple camera views and intervention environments to resolve partial observability and improve identifiability. 3) Temporal causal dynamics: Learn causal transitions and latent causal structure over time (including continuous-time dynamics) and enable multi-step rollouts. 4) Counterfactual interface: Provide APIs for do/soft/hard interventions in latent space and evaluate counterfactual validity. 5) Planning and transfer: Integrate the world model into partner-relevant planning/control pipelines and demonstrate sim-to-real transfer on representative manipulation tasks. 6) Tooling and transfer: Deliver reusable datasets, benchmarks, evaluation scripts, documentation, and a partner-ready software package.
R&D Target Specifications	**Target specifications for TRL 2 → TRL 4 prototype validation** • Latent state recovery (simulation): estimate robot joint angles and gripper state from RGB with mean per-joint MSE ≤ 0.05 rad² and average MCC ≥ 0.90 on held-out trajectories. • Robustness: maintain ≤ 2× degradation under unseen corruptions (e.g., 32×32 pixel occlusions, lighting/background randomization). • Multi-view observability: demonstrate accurate estimation of out-of-plane joints and object pose improvements when adding a second calibrated camera (frontal + overhead / side). • Temporal prediction: achieve stable K-step (e.g., K=10) latent rollouts with bounded error growth on pick-and-place trajectories; report success via task-level metrics (e.g., cube pose error, grasp success prediction). • Counterfactual validity: interventions in latent space produce qualitatively and quantitatively correct downstream effects (e.g., holding θ_{t+1} fixed stalls arm motion while object dynamics evolve as expected in simulation). • Planning impact: latent-space planning/policy learning achieves ≥20% absolute improvement in task success rate or ≥2× sample-efficiency over a comparable pixel-space baseline on partner tasks. • Runtime: ≥30 Hz inference on partner-selected hardware configuration (including encoder + dynamics rollout for short horizons).

R&D Plan	**Overall approach** The joint R&D is structured into three technical work packages and one transfer/commercialization work package, aligned with TIPA's two-phase program structure: • Phase 1 (6-month Pre-Study): Feasibility demonstrations, dataset design, initial prototypes, and commercialization alignment. • Phase 2 (2.5-year Main Research): TRL-4 prototype development, simulator + hardware validation, benchmarking, and transfer-ready packaging. **Work Package 1 (WP1):** Score-Based Interventional CRL for Robot & Scene State WP1 Objective (Evaluator-Facing) Develop a label-efficient causal state encoder that recovers controllable robot state and key scene variables from RGB observations by exploiting actuator-driven interventions. **WP1.1** – Interventional data pipeline Design intervention protocols (single-joint and multi-factor) in simulation-first digital twins (e.g., Panda-class manipulators) and extend to partner robots; support do/hard/soft interventions where feasible. **WP1.2** – Score-difference estimation and encoder learning Implement score-based objectives that exploit sparsity of score differences under interventions; operationalize via density-ratio estimation (binary classifiers) and/or diffusion/score models. **WP1.3** – Disentangled latent state Extend the ROPES-style two-stage autoencoder approach (dimensionality reduction + disentangling latent) from joint angles to joint+object state; support incremental disentanglement when only a subset of interventions is available. SME Value: Provides a foundation for robust perception and state estimation with drastically reduced labeling cost. Work Package 2 (WP2): Hybrid Interventional + Multi-View CRL **WP2 Objective** (Evaluator-Facing) Achieve full observability and improved identifiability by aligning latent representations across multiple camera views and intervention environments. **WP2.1** – Multi-view instrumentation Deploy synchronized multi-camera setups (e.g., wrist + external / frontal + overhead) with calibration and time alignment; build view-specific encoders with shared latent space. **WP2.2** – Score-consistency objectives Train encoders so latent estimates exhibit score consistency across both views and interventions, enabling recovery even with noisy or partially entangled views. **WP2.3** – Causal structure recovery Aggregate aligned latent estimates to recover the causal graph among latent factors using score-variation statistics.

SME Value: Improves deployability where single-view sensing is insufficient.

Work Package 3 (WP3): Temporal CRL for Causal Dynamics and Counterfactual Rollouts

WP3 Objective (Evaluator-Facing)

Learn causal dynamics in latent space and enable counterfactual rollouts for planning and risk assessment.

WP3.1 – Temporal identifiability and dynamics learning

Leverage temporally structured variation as implicit soft interventions; learn causal transitions and time-varying structure, including change-point detection where needed.

WP3.2 – Continuous-time modeling

Model smooth sensorimotor dynamics using continuous-time latent dynamics (neural ODE/SDE) with score-based constraints.

WP3.3 – Counterfactual API

Implement intervention operators over latent variables and quantify counterfactual validity on scripted trajectories (e.g., pick-and-place with variable mass/friction and hidden confounders).

SME Value: Enables safe what-if testing and faster iteration on control strategies without repeated physical trials.

Work Package 4 (WP4): Integration, Benchmarking, and Technology Transfer

WP4 Objective (Evaluator-Facing)

Deliver a TRL-4 validated prototype and partner-ready toolkit integrated with robotics workflows.

WP4.1 – Planning/control integration

Integrate latent world model into MPC / RL / Vision-Language-Action pipelines; demonstrate acceleration relative to pixel-space planning by operating on low-dimensional causal state.

WP4.2 – Benchmark suite and evaluation framework

Release datasets, scripts, and metrics for state recovery, OOD robustness, counterfactual validity, and policy transfer (simulation → real robot).

WP4.3 – Transfer and commercialization roadmap

Provide documentation, stable APIs, partner training, and an IP/commercialization plan for deployment in SME products.

SME Value: Reduces adoption friction and de-risks commercialization through validated benchmarks and integration assets.

Timeline and milestones aligned to TIPA phases

Phase 1, Pre-study and evaluation, July 1 to December 31, 2026

Phase 1 Objective:

Establish feasibility of interventional and multi-view causal state learning on a representative robot simulator, and finalize the detailed Phase-2 execution plan and partner integration requirements.

Phase 1 Milestones and Deliverables (by December 31, 2026):

• Dataset + intervention protocol specification (simulation-first)

• Prototype v0 causal state encoder (interventional CRL) with initial evaluation

• Multi-view alignment proof-of-concept

• Phase-2 detailed R&D plan, benchmarks, and integration interfaces

• Preliminary IP and commercialization pathway

Phase 2, Main research and project management, January 1, 2027 to June 30, 2029

2027 – Core Representation and Multi-view Learning

2027 H1: WP1 interventional CRL encoder v1; initial object state extension

2027 H2: WP2 hybrid multi-view alignment + causal structure recovery v1

Milestone: validated causal state recovery under environment randomization and occlusions.

2028 – Temporal Dynamics and Counterfactual Rollouts

2028 H1: WP3 temporal causal dynamics v1; continuous-time latent dynamics prototype

2028 H2: counterfactual API + evaluation; integrate with planning on scripted manipulation tasks

Milestone: demonstrated counterfactual validity and improved planning sample efficiency in simulation.

2029 – Sim-to-Real Transfer and Technology Transfer

2029 H1: full toolkit packaging, documentation, stable APIs; partner pilot on real robot

2029 H2: TRL-4 validation report, benchmark release, and commercialization package

Final Deliverables:

• TRL-4 validated CausalWorld prototype toolkit

• Datasets + benchmarks + evaluation scripts

• Integration guidance for partner robotics stack

• Commercialization and IP roadmap for partner-led deployment

Project Management and Review Structure

Quarterly joint steering committee meetings; monthly technical synchronization; structured risk management (data availability, sensor calibration drift, runtime constraints, and sim-to-real gaps).

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	[Research Institute] will lead the core algorithmic innovation and prototype development of CausalWorld, ensuring scientific rigor, theoretical grounding, and TRL progression. Responsibilities include: • Score-based CRL method design: interventional, multi-view, and temporal objectives; identifiability analysis where applicable. • Prototype software: implement the CausalWorld toolkit (encoders, dynamics, counterfactual API), with modular interfaces and documentation. • Benchmarking and evaluation: design datasets, tasks, and metrics (state recovery, OOD robustness, counterfactual validity, planning impact). • Technology transfer support: reference implementations, training materials, and joint debugging/integration with the partner. TIPA Alignment: Delivers globally competitive AI/robotics technology and systematically transfers know-how to the Korean SME partner.
Expected R&R of Korean SME Partner	The Korean SME partner will lead industrial integration, validation realism, and commercialization, ensuring that CausalWorld evolves into a deployable and market-relevant robotics capability. Responsibilities include: • Provide representative robotics use cases: tasks, environments, sensing configurations, and performance constraints relevant to the partner's products. • Data and integration infrastructure: enable simulator and hardware data collection, multi-camera setup, calibration procedures, and ROS 2 integration. • Evaluation and pilot deployment: run partner-defined benchmarks, validate runtime and reliability constraints, and provide usability feedback. • Commercialization leadership: identify target markets (industrial automation, logistics, service robotics), run pilot engagements, and define go-to-market plans. TIPA Alignment: Ensures commercial pull, rapid adoption, and sustainable economic impact for Korean SMEs.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected technical outcomes** By the end of the project, the joint R&D effort will deliver: • A TRL-4 validated CausalWorld prototype capable of learning causal latent state and dynamics from multi-view RGB observations under interventions and temporal variation. • Demonstrated label-efficient recovery of controllable robot state (e.g., joint angles) via actuator-driven interventions, reducing dependence on large labeled datasets. • Robustness evidence under distribution shifts (lighting/background changes, occlusions) and improved sim-to-real policy transfer enabled by causal latent factors.

	• A counterfactual (what-if) simulation interface for intervening in latent space and predicting downstream effects for planning and risk assessment. • Reusable datasets, benchmarks, and evaluation scripts supporting repeatable measurement of state recovery, OOD robustness, counterfactual validity, and planning impact. **Commercialization potential and industry impact** CausalWorld addresses a growing industry need: enabling reliable, generalizable robot autonomy with lower data-labeling cost and safer iteration through hypothetical evaluation. Commercialization pathways for the Korean SME partner include: • Deployment as an internal robotics development tool to accelerate perception, planning, and sim-to-real transfer. • Packaging as a licensable software module (causal state estimation + counterfactual simulator) for robotics OEMs and system integrators. • Integration into digital-twin platforms and robotics middleware stacks as an AI layer for counterfactual evaluation and robust planning. Value to industry stakeholders: • Robotics manufacturers: faster iteration and better reliability across changing deployments. • Industrial automation and logistics: improved robustness and reduced downtime from distribution shifts. • AI/robotics tool vendors: differentiated causal modeling and counterfactual reasoning capabilities.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization